Program: **Diploma in Microelectronics**	
Course Code: **6491C**	Course Title: **RF MEMS**
Semester: **6**	Credits: **4**
Course Category: **Program Elective**	
Periods per week: **4 (L:4, T:0, P:0)**	Periods per semester: **60**

Course Objectives:

- To understand the principle of operation of MEMS switches
- To know the operation of micromachined inductors and capacitors
- To understand the operation of micromachined RF filters and phase shifters
- To get an awareness on micromachined antenna and packaging for RF MEMS

Course Prerequisites:

Topic	Course code	Course Title	Semester
Electronic components		Fundamentals of Electrical and Electronics Engineering	2
Microsensor, microactuator, micromachining, micro system packaging		Introduction to MEMS	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Explain the principle of operation of MEMS switches	15	Understanding
CO2	Understand the principle of operation of micromachined inductors and capacitors	14	Understanding
CO3	Understand the principle of operation of micromachined RF filters and phase shifters	15	Understanding
CO4	Explain the fundamentals of micromachined antenna and packaging for RF MEMS	14	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3					
CO2	3	3					
CO3	3	3					
CO4	3	3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

<table>
<tr><th>Module Outcomes</th><th>Description</th><th>Duration (Hours)</th><th>Cognitive Level</th></tr>
<tr><td>CO1</td><td colspan="3">Explain the principle of operation of MEMS switches</td></tr>
<tr><td>M1.01</td><td>Discuss the basics of switching</td><td>4</td><td>Understanding</td></tr>
<tr><td>M1.02</td><td>Explain the switches for RF and microwave applications</td><td>5</td><td>Understanding</td></tr>
<tr><td>M1.03</td><td>Describe the actuation mechanisms for MEMS devices</td><td>4</td><td>Understanding</td></tr>
<tr><td>M1.04</td><td>Understand the working of bistable micro relays</td><td>2</td><td>Understanding</td></tr>
<tr><td colspan="4">Contents:
RF MEMS switches and micro relays (Basic theory only)
Introduction, Switch parameters, Basics of switching - Mechanical switches, Electronic switches, Switches for RF and microwave applications - Mechanical RF switches, PIN diode RF switches, Metal oxide semiconductor field effect transistors and monolithic microwave integrated circuits, RF MEMS switches, Actuation mechanisms for MEMS devices - Electrostatic switching, Mercury contact switches, Electromagnetic switching, Thermal switching, Bistable micro relays</td></tr>
<tr><td>CO2</td><td colspan="3">Understand the principle of operation of micromachined inductors and capacitors</td></tr>
<tr><td>M2.01</td><td>Discuss the basics of MEMS/micromachined passive elements</td><td>2</td><td>Understanding</td></tr>
<tr><td>M2.02</td><td>Explain various MEMS inductors</td><td>7</td><td>Understanding</td></tr>
<tr><td>M2.03</td><td>Understand the operation of MEMS capacitors</td><td>5</td><td>Understanding</td></tr>
<tr><td></td><td>Series Test - I</td><td>1</td><td></td></tr>
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<table>
<tr><td colspan="4">Contents:
MEMS inductors and capacitors (Basic theory only)
Introduction, MEMS/micromachined passive elements: pros and cons, MEMS inductors - Self-inductance and mutual inductance, Micromachined inductors, Folded inductors, Variable inductors, Polymer-based inductors
MEMS capacitors - MEMS gap-tuning capacitors, MEMS area-tuning capacitors, Dielectric tunable capacitors.</td></tr>
<tr><td>CO3</td><td colspan="3">Understand the principle of operation of micromachined RF filters and phase shifters</td></tr>
<tr><td>M3.01</td><td>Discuss various micromechanical filters</td><td>4</td><td>Understanding</td></tr>
<tr><td>M3.02</td><td>Understand the basics of surface acoustic wave filters</td><td>4</td><td>Understanding</td></tr>
<tr><td>M3.03</td><td>Describe the operation of MEMS phase shifters</td><td>4</td><td>Understanding</td></tr>
<tr><td>M3.04</td><td>Discuss the working of ferroelectric phase shifters</td><td>3</td><td>Understanding</td></tr>
<tr><td colspan="4">Contents:
Micromachined RF filters and phase shifters (No design required)
Micromechanical filters - Electrostatic comb drive, Micromechanical filters using comb drives, Micromechanical filters using electrostatic coupled beam structures
Surface acoustic wave filters - Basics of surface acoustic wave filter operation, Surface acoustic wave devices: capabilities, limitations and applications, Bulk acoustic wave filters, Micromachined filters for millimeter wave frequencies
Phase shifters - Types of phase shifters and their limitations
MEMS phase shifters - Switched delay line phase shifters, Distributed MEMS phase shifters, Polymer-based phase shifters
Ferroelectric phase shifters - Distributed parallel plate capacitors, Bilateral interdigital phase shifters, Interdigital capacitor phase shifters</td></tr>
<tr><td>CO4</td><td colspan="3">Explain the fundamentals of micromachined antenna and packaging for RF MEMS</td></tr>
<tr><td>M4.01</td><td>Discuss the fundamentals of micromachined antenna</td><td>6</td><td>Understanding</td></tr>
<tr><td>M4.02</td><td>Explain the packaging for RF MEMS</td><td>4</td><td>Understanding</td></tr>
<tr><td>M4.03</td><td>Understand the reliability issues in packaging</td><td>4</td><td>Understanding</td></tr>
</table>

	Series Test - I	1	

Contents:

Micromachined antennae (Basic theory only)

Overview of microstrip antennae - Basic characteristics of microstrip antennae, Micromachining techniques to improve antenna performance, Micromachining as a fabrication process for small antennae - Horn antenna for W-band, Tapered slot antenna

Micromachined reconfigurable antennae - Vee antenna

Packaging for RF MEMS devices and Reliability Issues

Role of MEMS packages - Mechanical support, Electrical interface, Protection from the environment, Thermal considerations

Types of MEMS packages - Metal, Ceramic, Plastic, Multilayer, Embedded overlay, Wafer-level, Microshielding and self-packaging

RF MEMS Packaging : Reliability issues - Packaging materials, Integration of MEMS devices with microelectronics, Wiring and interconnections, Reliability and key failure mechanisms

Text /Reference:

T/R	Book Title/Author
T1	Vijay Varadan, K. J. Vinoy, K. A. Jose, "RF MEMS and Their Applications", Wiley, 2003
R1	Gabriel M. Rebeiz, "RF MEMS: Theory, Design, and Technology", Wiley, 2003
R2	Hector J. De Los Santos,."RF MEMS Circuit Design for Wireless Applications", Artech House, 2002
R3	Eun Sok kim "Fundamentals of Micro electro mechanical Systems (MEMS)" McGraw Hill

Online resources:

Sl. No.	Website Link
1	https://omidi.iut.ac.ir/SDR/2008/Projects/Fereydani_SDR_project/References/%5B5%5D%20RF%20MEMS%20And%20Their%20Applications.pdf
2	https://resources.pcb.cadence.com/blog/2023-rf-mems-micro-electro-mechanical-systems-introduction
3	https://www.analog.com/en/product-category/mems-switches.html#category-detail
4	https://www.everythingrf.com/community/what-are-rf-mems